

CS3940 - Professional Portfolio

Heart Disease Prediction Project Report

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1. Executive Summary

The Heart Disease Prediction project leverages machine learning techniques to predict the likelihood of heart disease in patients based on various medical factors. Utilizing logistic regression, the model achieved an accuracy of approximately **84%**. This report details the project's background, objectives, methodology, system design, implementation, and conclusions, providing a thorough examination of the process and results.

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3. Introduction

3.1. Background

Heart disease remains one of the leading causes of death worldwide, responsible for millions of deaths each year. Early prediction and diagnosis are critical for effective treatment and prevention. The advent of machine learning has opened new avenues for predictive analytics in healthcare, offering the potential to identify at-risk individuals and provide timely interventions. This project utilizes machine learning techniques to develop a predictive model for heart disease using a dataset comprising various medical parameters.

3.2. Objectives

The primary objective of this project is to develop a robust machine learning model to predict heart disease. The specific objectives are:

- To preprocess and clean the dataset for optimal model performance.
- To explore and compare various machine learning algorithms to identify the most effective one.
- To develop a user-friendly interface for heart disease prediction that can be used by healthcare professionals and individuals alike.

3.3. Project Scope

The scope of the project encompasses data preprocessing, exploratory data analysis (EDA), feature engineering, model training, evaluation, deployment, and the development of a Streamlit-based user interface. The project is limited to the features provided in the dataset, focusing on logistic regression due to its initial superior performance. Additionally, other algorithms will be explored for comparative analysis.

3.4. Significance

Accurate prediction of heart disease can significantly improve patient outcomes by enabling early intervention and treatment. This project demonstrates the application of machine learning in healthcare, providing a framework that can be extended to other medical predictions. By making predictive tools accessible to healthcare providers, we can enhance decision-making processes and improve patient care.

4. Problem Statement

Heart disease prediction involves analyzing various medical parameters to determine the likelihood of disease presence. The challenge is to develop a model that can accurately classify patients based on these parameters, provide reliable predictions, and generalize well to new data. This involves overcoming issues related to data quality, feature selection, and model performance.

5. Feasibility

5.1. Feasibility Study

The feasibility of this project was evaluated based on technical, economic, and operational factors.

Technical Feasibility:

The project is technically feasible due to the availability of necessary libraries (such as **Pandas**, **NumPy**, **Scikit-learn**, **Matplotlib**, and **Seaborn**) and tools for data processing and model development. Additionally, the dataset is publicly available, making data acquisition straightforward.

Economic Feasibility:

The project is economically feasible as it relies on open-source software, which eliminates the need for expensive proprietary tools. The computational requirements are modest, making it suitable for execution on standard hardware without the need for specialized infrastructure.

Operational Feasibility:

The project is operationally feasible as it can be integrated into existing healthcare systems. The user-friendly interface developed using Streamlit allows for easy interaction, making it accessible to both healthcare professionals and patients.

5.2. SWOT Analysis

1. Strengths:

- High accuracy achieved by the logistic regression model.
- Interpretability of the logistic regression model, making it easy for healthcare professionals to understand and trust the predictions.
- Significant potential impact on healthcare by enabling early diagnosis and intervention.

2. Weaknesses:

- Limited to the features provided in the dataset, which may not capture all relevant aspects of heart disease risk.
- Potential for overfitting, especially if the model is overly complex relative to the dataset size.

3. Opportunities:

- Expansion to other medical predictions, such as diabetes or cancer risk assessment.
- Integration with electronic health records (EHR) for real-time prediction and decision support.
- Collaboration with healthcare providers to refine and validate the model.

4. Threats:

- Data privacy concerns, particularly with sensitive medical information.
- Potential model bias due to imbalanced datasets or unrepresentative samples.
- Regulatory challenges related to the deployment of predictive models in clinical settings.

5.3. Related Work

Previous studies have employed various machine learning techniques for heart disease prediction. Common algorithms include decision trees, neural networks, support vector machines (SVM), and ensemble methods like random forests and gradient boosting. This project builds upon these studies by focusing on logistic regression for its interpretability and effectiveness while exploring other algorithms for comparison.

6. Requirements Specification

6.1. Functional Requirements

- **Data Preprocessing:** Clean and normalize the dataset, handle missing values, and encode categorical variables.
- **Exploratory Data Analysis (EDA):** Perform EDA to understand the data distribution and relationships between features.
- **Feature Engineering:** Select and transform features to improve model performance.
- **Model Training:** Implement and train the logistic regression model and compare it with other machine learning algorithms.
- **Evaluation:** Assess model performance using metrics such as accuracy, precision, recall, and F1-score.
- **Model Deployment:** Save the trained model and develop a deployment pipeline.
- **User Interface:** Design and implement a Streamlit-based interface for user input and prediction display.

6.2. Non-Functional Requirements

- **Performance:** The model should provide predictions quickly and efficiently.
- **Usability:** The interface should be intuitive and user-friendly.
- **Security:** Ensure data privacy and secure handling of user inputs.
- **Scalability:** The system should be scalable to handle an increasing number of users and data volume.

7. System Design and Architecture

7.1. System Overview

The system consists of several components working together to provide accurate heart disease predictions. These components include data preprocessing, exploratory data analysis, feature engineering, model training, evaluation, deployment, and a user interface.

7.2. System Components

- Data Preprocessing Module: Handles data cleaning, normalization, and encoding of categorical variables. This module ensures that the data is in a suitable format for model training.
- Exploratory Data Analysis Module: Provides insights into the data distribution and relationships between features through visualizations and statistical analyses.
- Feature Engineering Module: Selects and transforms features to enhance model performance. This includes techniques such as feature scaling, interaction terms, and dimensionality reduction.
- Model Training Module: Implements and trains machine learning models, including logistic regression and other algorithms for comparison.
- Evaluation Module: Assesses model performance using various metrics and validation techniques. This module ensures that the model is reliable and generalizes well to new data.
- Model Deployment Module: Saves the trained model and develops a deployment pipeline for real-time predictions.
- User Interface Module: Built using Streamlit, this module provides an interactive interface for users to input data and receive predictions.

8. Implementation

Dataset Acquisition

The dataset was obtained from the UCI Machine Learning Repository. It includes 1024 instances with 14 attributes related to heart disease diagnosis. The dataset is split into features (X) and the target variable (Y), which indicates the presence or absence of heart disease.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Load the dataset
data = pd.read_csv('heart.csv')

# Check dataset shape and information
print(data.shape)
print(data.info())

# Display basic statistics
print(data.describe())
```

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
count	1025.000000	1025.000000	1025.000000	1025.000000	1025.000000	1025.000000	1025.000000	1025.000000	1025.000000	1025.000000
mean	54.434146	0.695610	0.942439	131.611707	246.000000	0.149268	0.529756	149.114146	0.336585	1.071000
std	9.072290	0.460373	1.029641	17.516718	51.592510	0.356527	0.527878	23.005724	0.472772	1.175000
min	29.000000	0.000000	0.000000	94.000000	126.000000	0.000000	0.000000	71.000000	0.000000	0.000000
25%	48.000000	0.000000	0.000000	120.000000	211.000000	0.000000	0.000000	132.000000	0.000000	0.000000
50%	56.000000	1.000000	1.000000	130.000000	240.000000	0.000000	1.000000	152.000000	0.000000	0.800000
75%	61.000000	1.000000	2.000000	140.000000	275.000000	0.000000	1.000000	166.000000	1.000000	1.800000
max	77.000000	1.000000	3.000000	200.000000	564.000000	1.000000	2.000000	202.000000	1.000000	6.200000

Data Preprocessing

Data preprocessing involved handling missing values, normalizing features, and encoding categorical variables. This step ensures that the model receives clean and standardized data for training.

```
data = data.dropna()
```

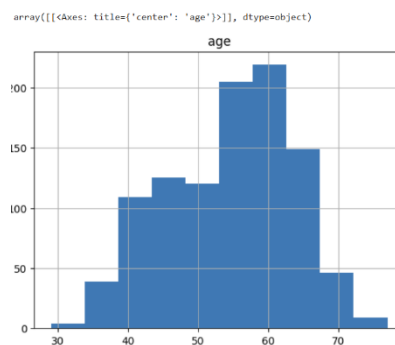
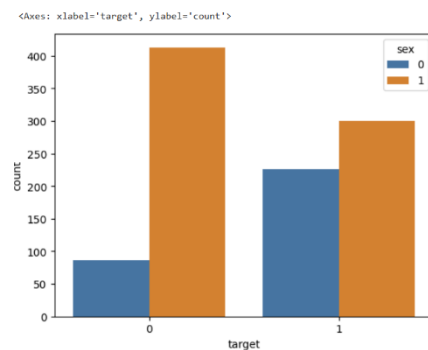
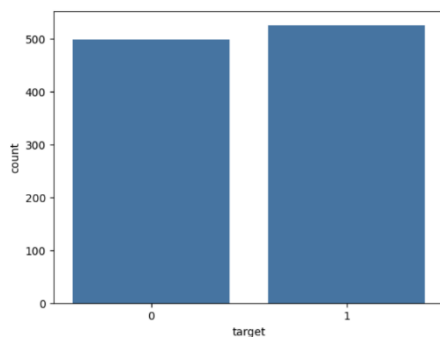
Exploratory Data Analysis (EDA)

EDA helps in understanding the distribution and relationships of the features. Visualizations were used to identify patterns and insights in the dataset.

```
# Visualize the distribution of target variable
sns.countplot(x='target', data=data)
plt.title('Distribution of Target Variable')
plt.show()

# Visualize the target variable by sex
sns.countplot(x='target', hue='sex', data=data)
plt.title('Target Variable by Sex')
plt.show()

# Histogram of age
data['age'].hist()
plt.title('Age Distribution')
plt.xlabel('Age')
plt.ylabel('Frequency')
plt.show()
```



Feature Engineering

Feature engineering involves selecting and transforming features to improve model performance. In this project, the features provided in the dataset were used directly, with potential transformations applied as necessary.

```
# Shuffle the data
data = data.sample(frac=1).reset_index(drop=True)

# Split the dataset into features and target variable
X = data.drop('target', axis=1)
Y = data['target']
```

Model Training

Logistic regression was chosen as the primary model due to its simplicity and effectiveness for binary classification tasks. The dataset was split into training and testing sets to evaluate the model.

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

# Split the dataset into training and testing sets
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.25, stratify=Y, random_state=2)

# Initialize and train the Logistic regression model
log_model = LogisticRegression()
log_model.fit(X_train, Y_train)

# Check accuracy on training data
Train_pred = log_model.predict(X_train)
train_acc = accuracy_score(Y_train, Train_pred)
print("Accuracy on training data = ", train_acc)

# Predict on test data and evaluate
predictions = log_model.predict(X_test)
test_acc = accuracy_score(Y_test, predictions)
print("Accuracy on test data = ", test_acc)
```

```
Accuracy on training data = 0.8372395833333334
Accuracy on test data = 0.8443579766536965
```

Model Evaluation

The logistic regression model achieved an accuracy of 84% on the test dataset, demonstrating good generalization performance. Further metrics such as precision, recall, and F1-score were calculated to provide a comprehensive evaluation.

```
from sklearn.metrics import classification_report  
  
# Generate classification report  
print(classification_report(Y_test, predictions))
```

Model Deployment

The trained logistic regression model was saved using **joblib** for deployment. A deployment pipeline was developed to allow real-time predictions.

```
import joblib  
  
# Save the model  
joblib.dump(log_model, 'logistic_regression_model.pkl')
```

User Interface Development

A Streamlit-based interface was developed to allow users to input data and receive predictions. The UI provides a simple and interactive way to use the model.

```
# Load the model
model = joblib.load('logistic_regression_model.pkl')

# Streamlit app
st.title("Heart Disease Prediction")
st.sidebar.header("Patient Data")

def user_input_features():
    age = st.sidebar.slider('Age', 20, 80, 50)
    sex = st.sidebar.selectbox('Sex', (0, 1))
    cp = st.sidebar.selectbox('Chest Pain Type', (0, 1, 2, 3))
    trestbps = st.sidebar.slider('Resting Blood Pressure', 90, 200, 120)
    chol = st.sidebar.slider('Serum Cholesterol', 100, 400, 200)
    fbs = st.sidebar.selectbox('Fasting Blood Sugar > 120 mg/dl', (0, 1))
    restecg = st.sidebar.selectbox('Resting ECG Results', (0, 1, 2))
    thalach = st.sidebar.slider('Maximum Heart Rate Achieved', 60, 200, 120)
    exang = st.sidebar.selectbox('Exercise Induced Angina', (0, 1))
    oldpeak = st.sidebar.slider('ST Depression', 0.0, 5.0, 2.5)
    slope = st.sidebar.selectbox('Slope of the Peak Exercise ST Segment', (0, 1, 2))
    ca = st.sidebar.selectbox('Number of Major Vessels Colored by Fluoroscopy', (0, 1, 2, 3))
    thal = st.sidebar.selectbox('Thalassemia', (0, 1, 2))

    data = {'age': age, 'sex': sex, 'cp': cp, 'trestbps': trestbps, 'chol': chol,
           'fbs': fbs, 'restecg': restecg, 'thalach': thalach, 'exang': exang,
           'oldpeak': oldpeak, 'slope': slope, 'ca': ca, 'thal': thal}
    features = pd.DataFrame(data, index=[0])
    return features

input_df = user_input_features()
st.write(input_df)

# Prediction
prediction = model.predict(input_df)
st.write("Prediction: ", "Heart Disease" if prediction == 1 else "No Heart Disease")
```

9. Conclusion

The **Heart Disease Prediction** project successfully developed a **logistic regression** model with an accuracy of **84%**. The project demonstrates the potential of machine learning in healthcare for early diagnosis and intervention. The user-friendly interface developed using **Streamlit** allows for easy interaction, making it accessible to both healthcare professionals and individuals. Future work can focus on expanding the feature set, exploring other algorithms, and enhancing the user interface for better accessibility.

10. Future Work

- User Interface Enhancement: Improve the user interface by adding more interactive features and visualizations to help users interpret the predictions better.
- Algorithm Comparison: Explore and compare more advanced machine learning algorithms, such as neural networks, ensemble methods, and deep learning techniques.
- Feature Expansion: Incorporate additional medical and demographic features to enhance model accuracy.

References

UCI Machine Learning Repository: Heart Disease Data Set. Available at: <https://archive.ics.uci.edu/ml/datasets/Heart+Disease>

Scikit-learn: Machine Learning in Python. Available at: <https://scikit-learn.org/>

Streamlit: The fastest way to build and share data apps. Available at: <https://streamlit.io/>

Pandas: Python Data Analysis Library. Available at: <https://pandas.pydata.org/>

Matplotlib: Visualization with Python. Available at: <https://matplotlib.org/>

Seaborn: Statistical Data Visualization. Available at: <https://seaborn.pydata.org/>